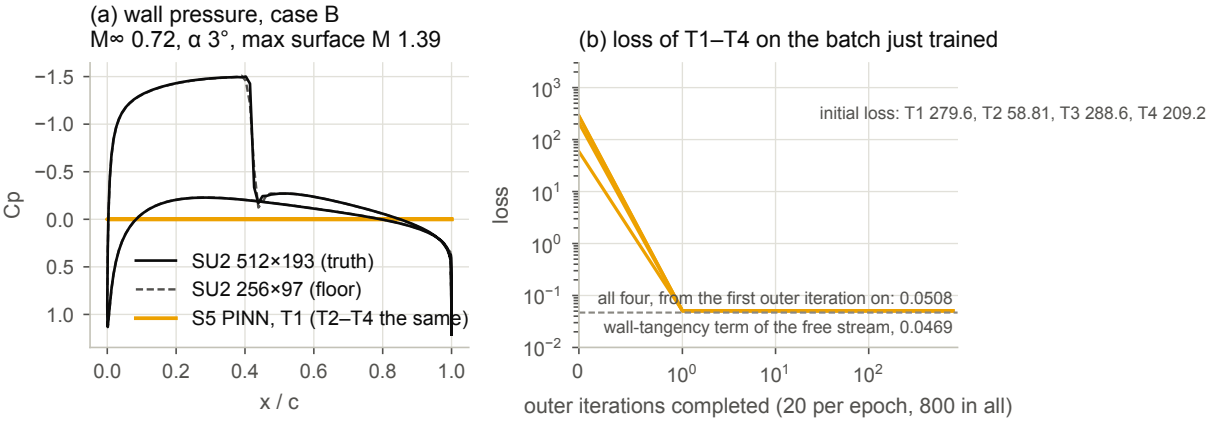
(a) wall pressure, case B
M∞ 0.72, α 3°, max surface M 1.39
Cp
−1.5
−1.0
−0.5
0.0
0.5
1.0
0.0
0.2
0.4
0.6
0.8
1.0
x / c
SU2 512×193 (truth)
SU2 256×97 (floor)
S5 PINN, T1 (T2–T4 the same)
(b) loss of T1–T4 on the batch just trained
loss
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
initial loss: T1 279.6, T2 58.81, T3 288.6, T4 209.2
all four, from the first outer iteration on: 0.0508
wall-tangency term of the free stream, 0.0469
0
10^{0}
10^{1}
10^{2}
outer iterations completed (20 per epoch, 800 in all)